An atom consists of a cloud of electrons surrounding a central, massive nucleus. The electron cloud is roughly 10^{-8} centimeter in size (about a millionth the diameter of a human hair), and the nucleus at its core is 100,000 times smaller, roughly 10^{-13} centimeter; see the diagram below. If the electron cloud were enlarged to the size of the Earth, then the nucleus would become the size of a football field. Despite its tiny size, the nucleus is several thousand times heavier than the tenuous electron cloud.

The negatively charged electrons are held in their cloud by the electrical pull of the positively charged nucleus, but they do not fall into the nucleus for the same reason as a white-dwarf star does not implode: A quantum mechanical law called the Pauli exclusion principle forbids more than two electrons to occupy the same region of space at the same time (two can do so if they have opposite "spins," a subtlety ignored in Chapter 4). The cloud's electrons therefore get paired together in cells called "orbitals." Each pair of electrons, in protest against being confined to its small cell, undergoes erratic, high-speed "claustrophobic" motions, like those of electrons in a white-dwarf star (Chapter 4). These motions give rise to "electron degeneracy pressure," which counteracts the electrical pull of the nucleus. Thus, one can think of the atom as a tiny white-dwarf star, with an electric force rather than a gravitational force pulling the electrons inward, and with electron degeneracy pressure pushing them outward.

The right-hand diagram below sketches the structure of the atomic nucleus, as discussed in the text; it is a tiny cluster of protons and neutrons, held together by the nuclear force.

